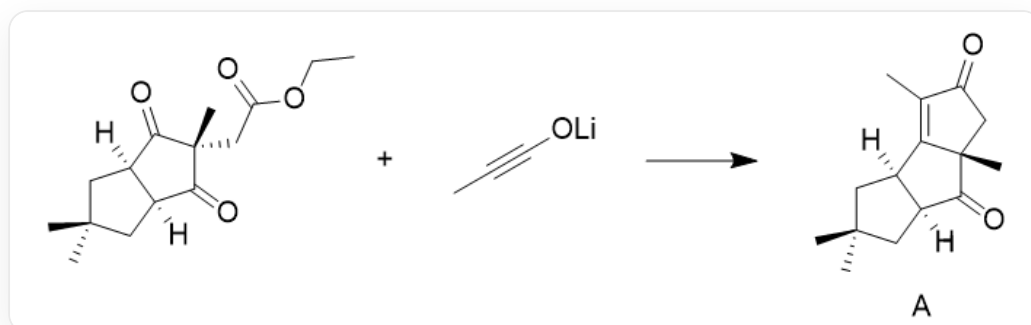


Question

Organolithium alkynolates are one of the commonly used organolithium reagents. As shown in the reaction below, pre-prepared lithium propargyl alcoholate can be used to synthesize the natural product **A** with another substrate. Two key electroneutral intermediates **B** and **C** are formed during the formation of **A**.



CC1(C)C[C@]2([H])[C@](C([C@@](C)(CC(OCC)=O)C2=O)=O)([H])C1和CC#CO[Li]反应可合成
CC1=C2[C@@](CC1=O)(C)C([C@@]3([H])CC(C)(C)C[C@]32[H])=O

The following statements are available:

1. **A** and **B** have the same number of rings.
2. **A** and **C** have the same number of rings.
3. **B** and **C** have the same number of oxygen atoms.
4. **B** contains an alkynyl group.
5. **C** contains 4 double bonds.
6. Gas is released during the formation of **A**.

Which of the above statements are correct:

A. 1, 4

B. 2, 3

C. 2, 4

D. 3, 5

E. 3, 6

F. 5, 6

G. 6

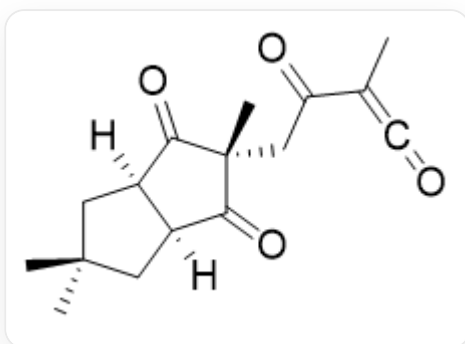
Answer

Correct Answer: **E**

Detailed Explanation

The reaction mainly proceeds through three steps to obtain **A**:

1. In lithium alkynol, the alkyne terminal carbon connected to oxygen carries a positive charge, and the other end carries a negative charge. During the reaction, the electrons of the oxygen anion are transferred to form a lithium enolate of ketene, which promotes the negatively charged terminal carbon to attack the ester carbonyl carbonyl in another reactant. Ethoxy leaves, and a nucleophilic substitution reaction occurs to generate intermediate product **B**.
- . The chemical formula of **B** is:



CC1(C)C[C@]2([H])[C@](C([C@@](C)(CC(C(C)=C=O)=O)C2=O)=O)([H])C1

CHECKPOINT

1 PTS

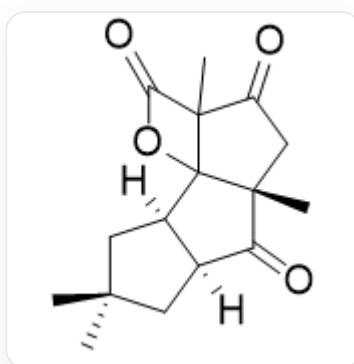
A nucleophilic substitution reaction occurs, and the negatively charged terminal carbon in the ketene anion attacks the ester carbonyl carbonyl of another reactant.

CHECKPOINT

2 PTS

The chemical formula of **B** is CC1(C)C[C@]2([H])[C@](C([C@@](C)(CC(C(C)=C=O)=O)C2=O)=O)([H])C1

2. There are two carbonyl groups on the ring in **B**. Among them, the carbonyl group located at the top of the figure undergoes a [2+2] cycloaddition reaction with the carbon-carbon double bond in the ketene to form compound **C** with a four-membered ring and a five-membered ring. The chemical formula of **C** is:



O=C1[C@@]2([H])CC(C)(C)C[C@@]2([H])C3(O4)[C@]1(C)CC(C3(C)C4=O)=O

CHECKPOINT

1 PTS

A [2+2] cycloaddition reaction occurs, and the carbonyl group in **B** reacts with the carbon-carbon double bond in the ketene to form product **C**.

CHECKPOINT

2 PTS

The chemical formula of **C** is O=C1[C@@]2([H])CC(C)(C)C[C@@]2([H])C3(O4)[C@]1(C)CC(C3(C)C4=O)=O

3. The ester group in **C** undergoes an elimination reaction, producing CO_2 gas and obtaining product **A** with a carbon-carbon double bond.

CHECKPOINT

1 PTS

An elimination reaction occurs, producing CO_2 gas and a carbon-carbon double bond

Therefore, the existing statements 3 and 6 are correct. Select option E.